

## **Distance and Delay**

### **How Commute Distance Shapes Waiting Behavior and Mode Choice Among MIT Students: An Analysis of Waiting Tolerance, Perceived Reliability, and Mobility Adaptation**

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## Highlights

- Longer commute distances are associated with stronger preferences for controllable modes (e.g., biking, scootering) over scheduled transit.
- Students who live farther away report more deliberate trip planning and reduced willingness to experience unplanned waiting.
- Perceived waiting tolerance varies significantly across mode groups, suggesting behavioral adaptation rather than uniform attitudes.
- Findings may inform targeted improvements to campus transit reliability and last-mile connectivity in support of decarbonized mobility.
- Survey-based insights demonstrate how micro-scale commuter behavior can guide broader sustainable transportation strategies.

## Abstract

Understanding how commute distance shapes travel behavior is critical for designing effective, low-carbon transportation systems. This study examines whether MIT students who live farther from campus exhibit greater tolerance for waiting and delays or instead avoid waiting by choosing faster, more controllable modes such as biking or scootering. Using an original survey of 75 MIT students, the research collects self-reported commute distance, primary mode choice, perceived wait tolerance, and attitudes toward transit reliability. Commute distance was self-reported through dorm selection or approximate distance estimates and categorized for analysis. Descriptive statistics, cross-tabulations, and correlation analysis are used to explore relationships between distance, waiting behavior, and mode choice.

Results indicate that commute distance meaningfully influences mode choice: students living farther from campus display a higher probability of choosing autonomous, time-controllable modes and a lower likelihood of relying on scheduled bus or shuttle services. Contrary to the possibility that longer-distance commuters develop greater patience, the findings suggest that they instead minimize exposure to waiting by selecting modes with low schedule dependence. Perceived waiting tolerance also varies across mode groups, with transit users reporting greater acceptance of delays than non-transit users, consistent with behavioral adaptation to the inherent characteristics of their chosen mode.

These insights have implications for campus mobility planning and broader decarbonization objectives. By understanding how distance affects waiting behavior, planners can design targeted reliability improvements, last-mile connections, and mode-shift incentives that better align with commuter needs. The study also highlights how micro-level behavioral evidence from campus settings can contribute to scalable sustainable mobility strategies. Rather than developing greater patience, students with longer commutes adapt by avoiding waiting altogether through the selection of autonomous, schedule-independent modes, indicating that waiting tolerance is shaped by mode choice rather than commute distance.

## **Problem Statement**

### **Research Questions**

- How does commute distance relate to students' tolerance for waiting and delays?
- Do students who live farther from campus compensate for longer travel times by choosing modes with more controllability (e.g., biking, scootering), rather than scheduled modes that involve waiting (e.g., shuttle, bus)?
- How do perceptions of reliability and planning flexibility vary with different commute distances and modes?

### **Hypotheses**

These hypotheses are designed to test whether distance leads students to adapt by becoming more patient or whether it motivates them to avoid situations involving waiting altogether.

- H1: Students with longer commutes will report lower tolerance for unplanned delays or will avoid waiting entirely through mode choice.
- H2: Longer-distance commuters will show a higher likelihood of selecting autonomous, schedule-independent modes (biking, scootering) over scheduled transit.
- H3: Transit users will exhibit higher waiting tolerance than non-transit users as a result of adaptation to scheduled, headway-dependent systems.
- H4: Mode choice may also be influenced by secondary factors such as weather, physical mobility constraints, or activity chaining, which may interact with distance effects.

### **Significance/Purpose**

College environments function as communities of urban mobility systems, where behavioral adaptation to distance and reliability can be observed at high resolution. MIT's environment of being dense, walkable, multimodal provides an ideal setting to study behavioral responses to waiting, perceived reliability, and modal controllability.

The findings can inform practical improvements to MIT's transportation system, such as targeted shuttle reliability enhancements, optimized headways, and last-mile connectivity strategies for students living off campus. Because waiting is a well-documented deterrent to public transit use, identifying which groups are most sensitive to delay can help planners allocate resources more effectively.

More broadly, this study connects to transportation decarbonization by identifying behavioral levers that support low-carbon mobility. Understanding when students choose biking or walking over ridehailing, or when reliable transit encourages mode shift away from private vehicles, can guide demand-side strategies for reducing emissions. If distance-driven waiting aversion pushes commuters toward sustainable modes, planners can reinforce that shift; if it pushes them away from transit, improving reliability may help preserve low-carbon mode share.

## **Literature Review**

This literature review synthesizes major strands of research relevant to this project: (a) waiting and perceived delay, (b) transit reliability and behavioral adaptation, (c) commute distance and mode choice, and (d) psychosocial factors such as control, predictability, and perceived autonomy. Together, these bodies of work provide the foundation for examining how MIT students' commute distances shape their waiting tolerance and mobility decisions.

### **a. Waiting Time and Delay Perception**

Waiting time consistently ranks among the most negatively perceived components of travel. Classical transportation economics treats waiting as a cost with a higher disutility weight than in-vehicle time. Wardman (2004) found that travelers often perceive waiting time to be worth 1.5-2.5x the disutility of in-vehicle time, a multiplier replicated in subsequent studies across Europe, Asia, and North America. Similarly, Small (2012) argues that uncertainty amplifies this negative perception, suggesting that unpredictable waiting contributes to disproportionately high dissatisfaction.

A key insight in behavioral research is that unplanned waiting is psychologically distinct from planned waiting. Fan, Guthrie, and Levinson (2016) demonstrate that passengers experience less stress and frustration when they plan around predictable service headways. This supports earlier work by Mazloumi et al. (2010), showing that reliability-more than average travel time-governs perceived quality of transit. When headways are long, passengers strategically adjust their behavior to minimize unstructured waiting, often arriving “just-in-time” or avoiding transit altogether.

These findings motivate the question in the current study: If waiting is highly aversive, do longer-distance commuters become more tolerant through repeated exposure, or do they choose modes that allow them to minimize waiting entirely?

### **b. Transit Reliability and Behavioral Adaptation**

Reliability is defined as consistency in travel time and expected arrival. Reliability plays a significant role in shaping user satisfaction and mode retention. Bates, Polak, Jones, and Cook (2001) found that travelers incorporate reliability into their generalized travel cost calculations, often overweighting variability compared to average travel time. More recent research has identified “buffer behaviors,” in which individuals add slack to their trips to compensate for potential delays (Watling, 2020). However, adaptation is not uniform: habitual transit riders develop coping strategies such as schedule checking, alternative route planning, or accepting delays as part of the mode's inherent characteristics.

Studies of college campuses mirror these patterns. Transportation systems at universities often rely on lower-frequency shuttle services, creating longer headways and less predictability than metropolitan transit networks. Zhao et al. (2014), studying campus shuttles in Beijing, found that students compensate for unreliable service through temporal flexibility and increased tolerance of small delays, which is consistent with long-term mode adaptation. However, they also found that students with access to bicycles were more likely to switch modes when delays exceeded their personal “acceptability threshold.”

This supports the hypothesis in the present research that waiting tolerance may be higher among current transit users due to adaptive behavior, rather than among long-distance commuters.

### **c. Distance, Travel Time, and Mode Choice**

Commute distance is a major determinant of travel mode selection, influencing the trade-offs travelers make between time, cost, reliability, and physical effort. McFadden’s (1974) discrete choice theory, later extended by Ben-Akiva and Lerman (1985), shows that distance interacts with generalized cost to structure mode preferences. Walking becomes less attractive as distances increase; biking becomes more competitive in the 1-3 mile range; transit and micromobility become dominant for moderate distances; and motorized modes dominate longer ranges.

Among college students, distance interacts with infrastructure availability. Bopp, Kaczynski, and Wittman (2011) found that students living farther from campus were significantly more likely to bike if safe cycling infrastructure was available, reflecting a desire for “door-to-door” control. Similarly, Sisson & Tudor-Locke (2008) observed that students living within one mile overwhelmingly walked, while those 1-3 miles away used a mix of biking and transit depending on weather and reliability. Those beyond 3 miles were more likely to rely on motorized modes due to time and comfort constraints.

These distance thresholds are important for the present study because MIT’s campus sits within dense, walkable Cambridge and integrates well with cycling infrastructure. Students living farther away may have stronger incentives to select modes that minimize waiting and maximize autonomy: particularly biking and scootering.

### **d. Control, Autonomy, and Predictability**

Transportation decisions are shaped not only by time and cost but also by perceived control. Kahneman and Tversky’s (1979) foundational work in behavioral economics argues that individuals avoid uncertainty and unpredictable losses, trends echoed in contemporary transit research. Lyons, Jain, and Holley (2007) found that travelers prefer modes that reduce stress by offering autonomy, even if travel times are slightly longer. This is consistent with studies showing higher user satisfaction with biking because of its predictability and independence from traffic or schedules (Heinen, van Wee, & Maat, 2010).

For college students in particular, autonomy is central. Shaheen and Cohen (2013) note that students often prefer modes that minimize waiting, maximize flexibility, and integrate well with irregular schedules. Scooters, bikes, and walking provide controllability, while shuttles and buses require accepting the risk of delay. The literature therefore suggests that students with longer commutes may gravitate toward modes with high autonomy to offset increased travel time and reduce psychological stress.

### **Literature Synthesis:**

The literature establishes three major foundations for this research:

1. Waiting time is disproportionately aversive, especially when unplanned.
2. Transit reliability shapes tolerance; current transit users often adapt and become more accepting of small delays.
3. Distance drives mode choice, with longer-distance commuters tending toward controllable, autonomous modes if available.

No existing study examines whether students with longer commutes specifically develop greater tolerance for waiting or whether they strategically avoid waiting by shifting modes. Nor has previous work combined commute distance, waiting tolerance, and mode choice among MIT students specifically, a population with unique transportation options (dense bike network, campus shuttles, compact geography). This study fills a gap by integrating behavioral, spatial, and perceptual data to understand how distance shapes the trade-off between waiting tolerance and mode controllability.

### **Method: Experiment Design or quasi-experiment design**

#### **Experiment Design**

Because this study investigates existing differences in students' commuting behavior rather than manipulating conditions, the project uses a quasi-experimental, natural-experiment design. MIT's student housing (on-campus, near-campus, and farther off-campus residences) naturally assigns individuals to different commute distances and mode options, creating variation without researcher intervention.

#### **Monitoring strategy: before condition and after condition**

Although no intervention was implemented, the study conceptually adopts a before/after monitoring analogy:

- "Before condition" = students' baseline commuting patterns: commute distance, travel mode, waiting tolerance, and commute time.
- "After condition" = observed behavioral outcomes or attitudes conditional on these baseline characteristics: preferred modes, avoidance of waiting, sensitivity to delay, and perceived flexibility.

The “before” and “after” framing allows us to interpret distance as an external condition shaping subsequent decisions, similar to a natural experiment where geography acts as the assignment mechanism.

### **Intervention, nudge, information**

No behavioral intervention, nudge, or informational treatment was administered. The study instead analyzes naturally occurring behavioral responses to commute distance, such as: self-selected mode choice, spontaneous avoidance of waiting-intensive modes, and adaptations (e.g., biking instead of taking a bus). This aligns with the principles of observational behavioral science where environmental conditions (distance) function as the “treatment.”

### **Data collection strategy: 1st hand data**

Data used in this study are first-hand, collected through an original survey distributed to MIT students.

**Technology:** Google Forms used to collect structured multiple-choice responses and short free-response entries.

**Instruments / implementation:** The survey instrument included questions across four domains:

1. Commute distance category (on-campus, short walk/bike, long walk, beyond walkable distance)
2. Primary commute mode (walk, bike, scooter, public transit, car)
3. Wait tolerance (numeric or short qualitative response: “max minutes I am willing to wait”)
4. Typical one-way commute time

The form was designed to take under 2 minutes, maximizing response rates. The survey was distributed via dormspam (dorm mailing lists).

### **iii. Sampling framework / Questionnaire design**

**Sampling approach:** This is a convenience sample of MIT students and should be treated as non-probability sampling. The sample likely overrepresents on-campus residents and students who actively check dorm email lists.

**Population:** individuals traveling regularly to the MIT campus.

**Goal:** capture variation in commuting strategies across diverse home locations.

The questionnaire followed two design principles:

- Simplicity: short, unambiguous multiple-choice items.
- Behavioral specificity: focus on costs, timing, and choices rather than preferences.

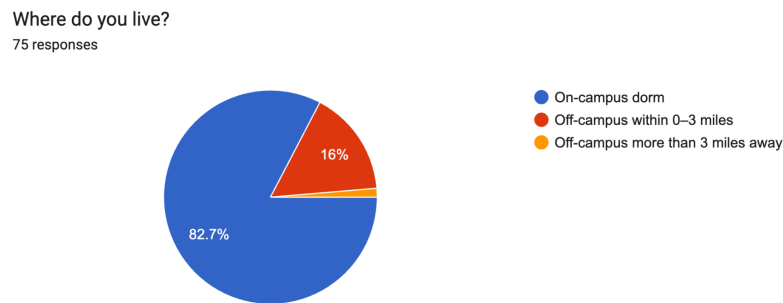
This study uses observational, survey-based data and a convenience sample. Commute distance functions as an externally determined condition, while mode choice and waiting tolerance are jointly determined. Results should therefore be interpreted as descriptive and hypothesis-generating rather than causal.

### **Full results and Interpretations**

## Sample Characteristics

A total of 75 MIT students completed the survey. As expected given the dorm-based distribution method, the sample is heavily weighted toward on-campus residents, with a smaller share of respondents living off campus within three miles of campus. One respondent reported living more than three miles away.

Due to the presence of only a single respondent in the >3 mile category, this group is excluded from comparative analysis and is discussed only descriptively. All distance-based comparisons therefore focus on two groups: on-campus residents and off-campus residents living within 0-3 miles of campus.



*Figure 1. Distribution of respondents by commute distance category*

Despite this imbalance, the sample includes sufficient variation across these two distance categories to examine differences in mode choice and waiting behavior. The >3 mile respondent primarily relies on a car for commuting and is included only to provide contextual completeness rather than as part of the analytical comparisons.

## Mode Choice Varies Strongly With Commute Distance



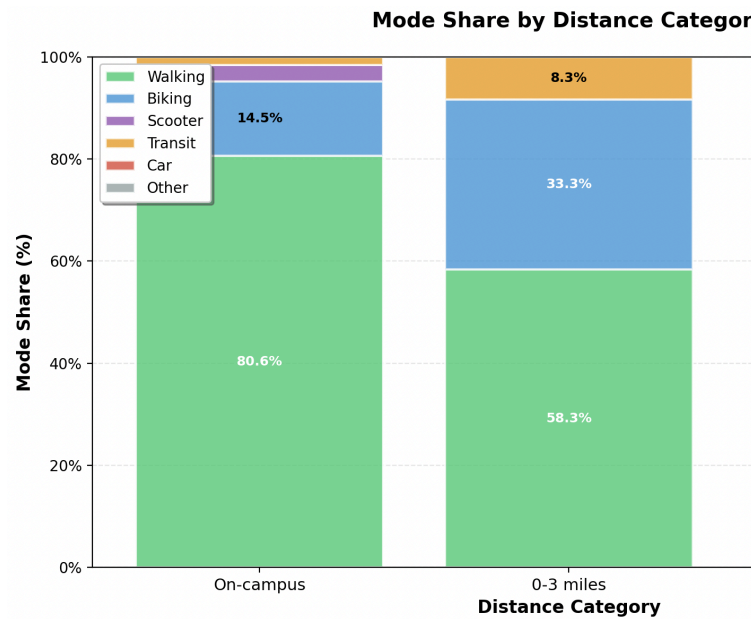


Figure 2. Primary commute mode by distance category

Mode choice differs substantially by commute distance category. Nearly all on-campus respondents report walking as their primary commute mode, reflecting the short travel distances and high walkability of campus residences. In contrast, students living off campus within 0-3 miles exhibit a pronounced shift toward biking and scooting.

This pattern is consistent with the hypothesis that longer commute distances incentivize the selection of autonomous, schedule-independent modes. Off-campus respondents overwhelmingly rely on biking or scooting, modes that minimize exposure to waiting and allow for flexible departure timing, rather than scheduled transit services that involve headway-based delays. Walking remains uncommon in this group due to increased distance, while reliance on buses or shuttles remains limited.

These results support H2 and suggest that commute distance shapes mobility behavior primarily through mode selection rather than through increased tolerance for waiting.

### Waiting Tolerance (Minutes)

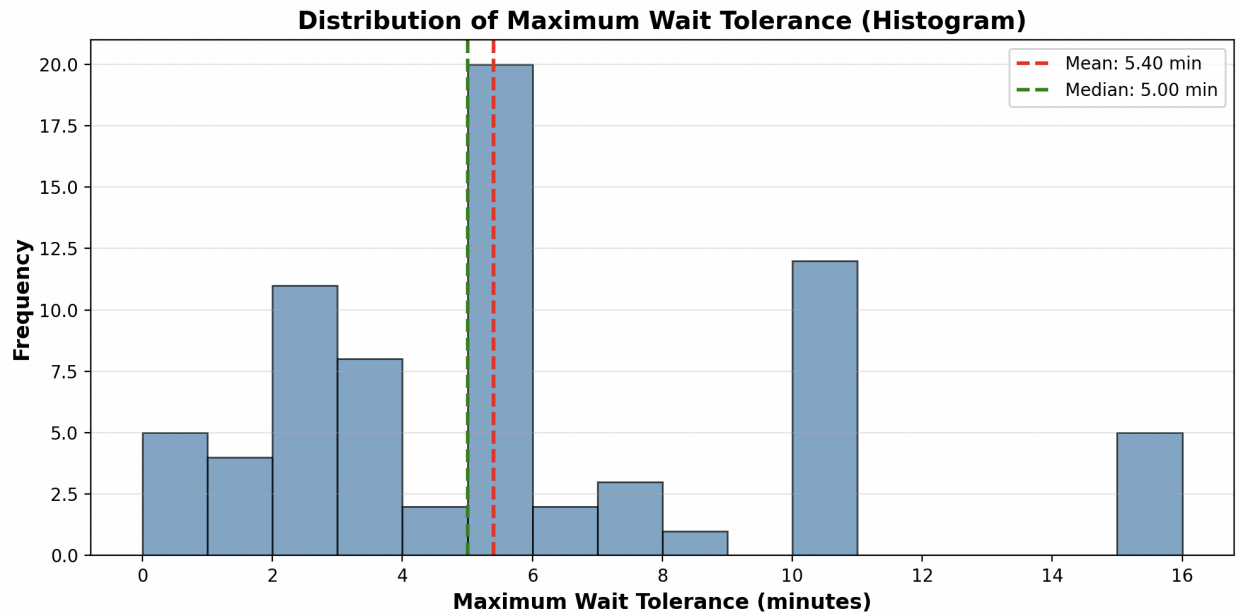


Figure 3. Distribution of maximum wait tolerance (minutes)

Students reported the maximum amount of time they would be willing to wait for a campus shuttle or MBTA bus before switching to another option. Reported waiting tolerances ranged from 0-15 minutes, with most responses clustered at the lower end of the distribution.

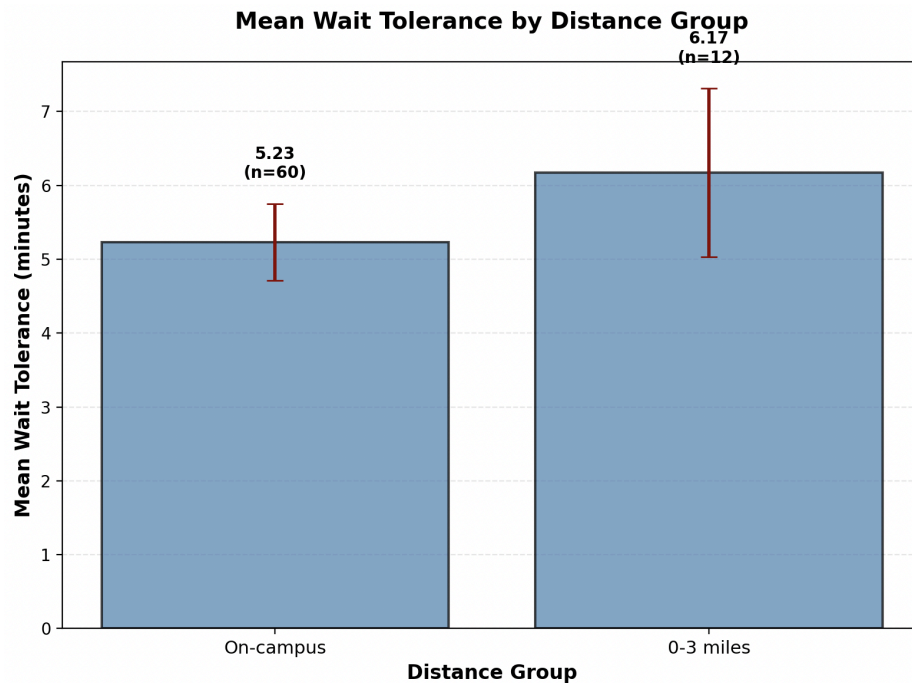


Figure 4. Mean wait tolerance by distance category

When comparing waiting tolerance across commute distance categories, mean tolerance increases only marginally between on-campus residents and off-campus residents living within 0-3 miles. This difference is small and not statistically meaningful, indicating that students living farther from campus do not exhibit substantially greater willingness to wait for transit.

This finding is consistent with H1 and suggests that longer commute distances do not lead students to develop greater patience for delays. Instead, distance appears to influence mobility behavior primarily through mode choice rather than through increased tolerance for waiting.

### Mode Choice Predicts Waiting Tolerance

Waiting tolerance varies substantially by primary commute mode. Students who primarily rely on scheduled transit services, such as campus shuttles or MBTA buses, report significantly higher tolerance for waiting compared to walkers, bikers, and scooter users. Transit users are the only group with consistent willingness to wait beyond 10 minutes, while non-transit users overwhelmingly report tolerance thresholds below five minutes.

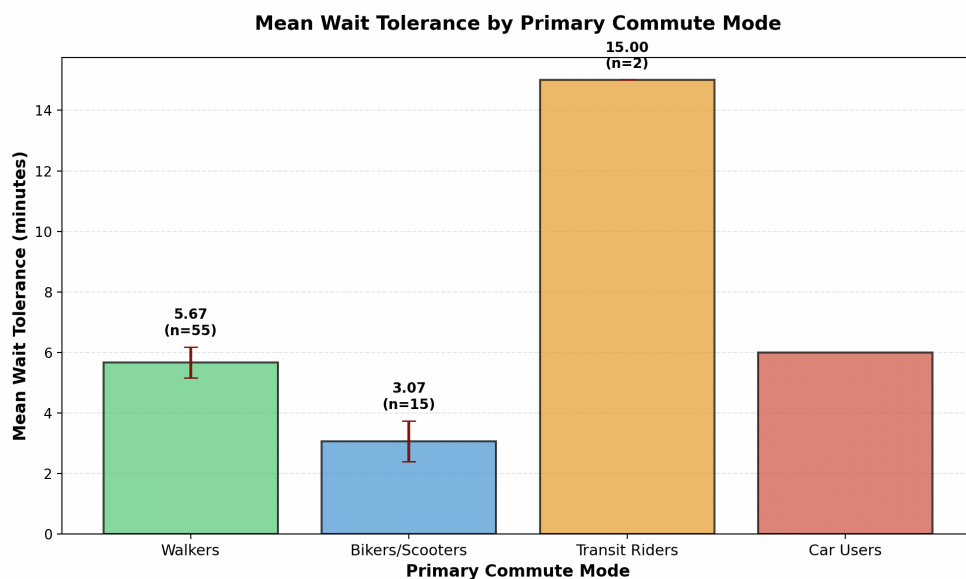


Figure 5. Mean wait tolerance by mode group

This pattern supports H3 and suggests that waiting tolerance reflects behavioral adaptation to headway-based systems rather than an intrinsic preference for waiting. Students who rely on transit appear to incorporate expected delays into their planning and adjust their expectations accordingly. In contrast, users of autonomous modes maintain high optionality and demonstrate low tolerance for unplanned waiting, consistent with their ability to depart at will and avoid schedule dependence.

Although the number of transit users in the sample is small, the consistency of this pattern across both numeric wait thresholds and self-rated tolerance measures reinforces the conclusion that mode choice, rather than commute distance, is the primary predictor of waiting tolerance.

### Behavioral Response to an 8-Minute Delay

Participants were asked what they would do if the bus were delayed by eight minutes. Most respondents chose to switch modes (usually walking or biking) rather than wait.

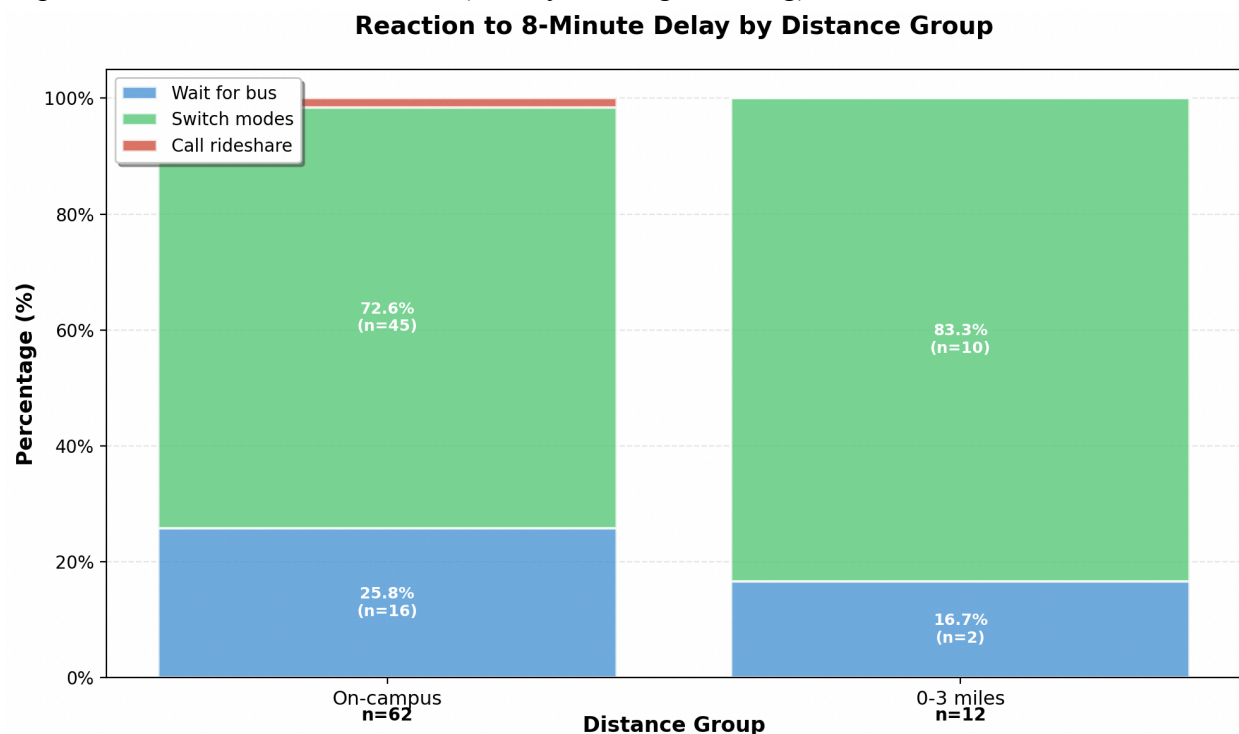


Figure 6. Responses to an 8-minute delay by distance category

Notably:

- On-campus residents predominantly chose to walk/bike (switch modes on chart) instead.
- Off-campus residents (0-3 miles) made similar choices, favoring walking/biking.

Transit riders were the only group consistently choosing to wait. This reinforces the finding that mode, not distance, predicts patience. Students who already rely on transit decide to wait because switching modes is not feasible at the moment, whereas walkers/bikers maintain high optionality and avoid waiting entirely.

### Commute Time and Distance Relationship

To assess whether the self-reported distance categories correspond to meaningful differences in travel experience, commute time was examined descriptively across distance measures.

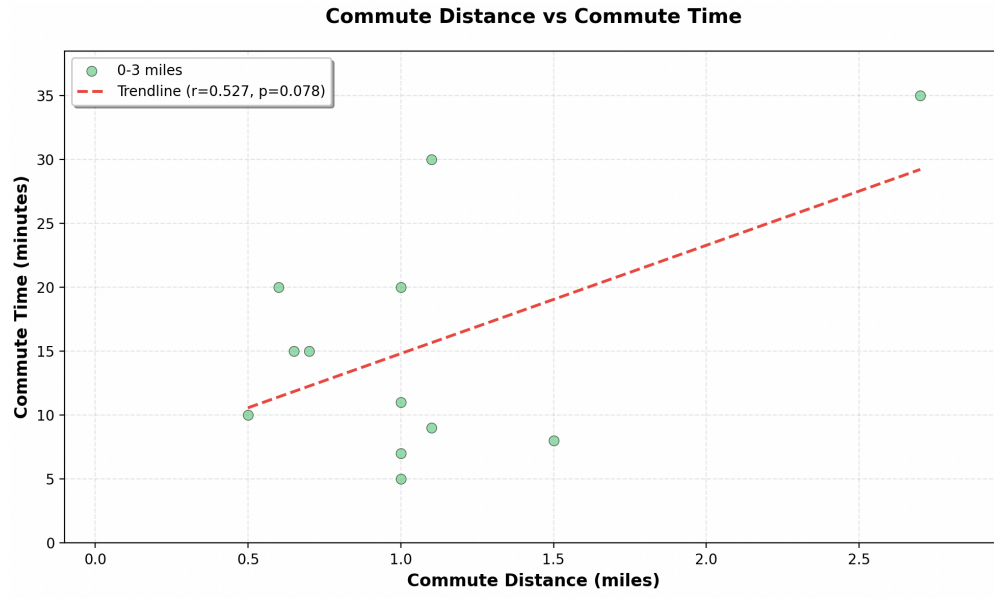


Figure 7. Commute distance (miles) vs. commute time (minutes)

The scatterplot shows a weak positive relationship: students living farther away experience longer travel times, even when using faster modes like biking. This suggests that distance is associated with differences in travel time and functions as a relevant contextual condition for interpreting mode choice. For even more support figure 8 is provided below:

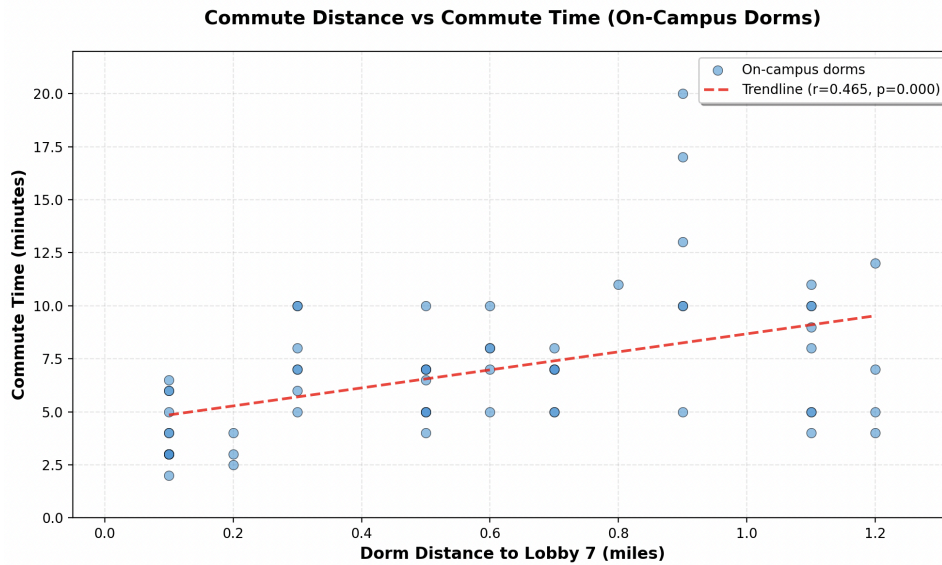


Figure 8. Commute distance (miles) vs. commute time (minutes) - On-Campus Dorms

The scatterplot shows a stronger positive relationship. 0.1 is designated the closest dorm to Lobby 7 (Maseeh), and 1.2 is the furthest (Tang). Order is as follows: Maseeh (closest): 0.1 miles, McCormick: 0.2 miles, EC (East Campus): 0.3 miles, Baker: 0.4 miles, New Vassar: 0.5 miles, BC (Burton-Conner): 0.6 miles, Number Six & Random Hall: 0.7 miles, MacGregor: 0.8 miles, Simmons: 0.9 miles, New House: 1.0 miles, Next House: 1.1 miles, Tang: 1.2 miles

(farthest). Though these are not the exact distances away by walk, it is a good relative measure to evaluate the relationship between distance and commute time to validate that distance creates travel differences.

### Self-Reported Waiting Tolerance (0-10 Scale)

Participants also rated their general tolerance for waiting on a 0-10 scale. Values ranged widely, but the majority fell between 2 and 6.

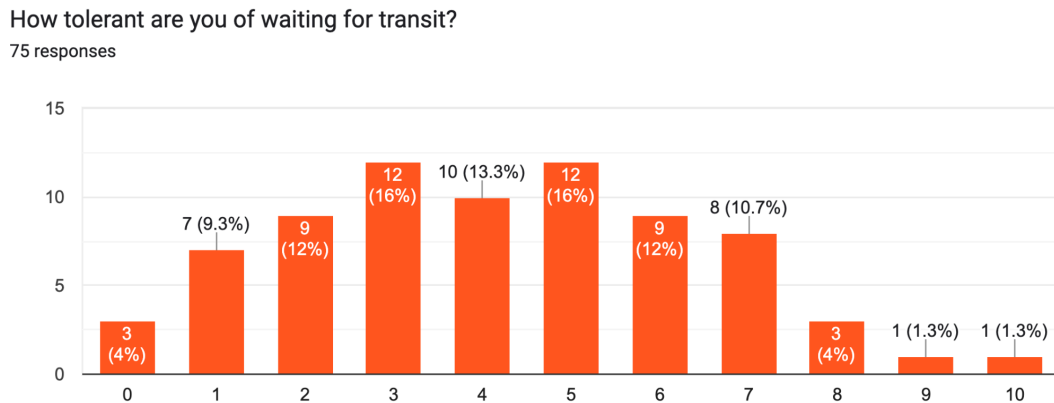


Figure 9. Distribution of waiting tolerance rating (0-10)

When broken down by mode, transit users scored higher than both walkers and bikers.

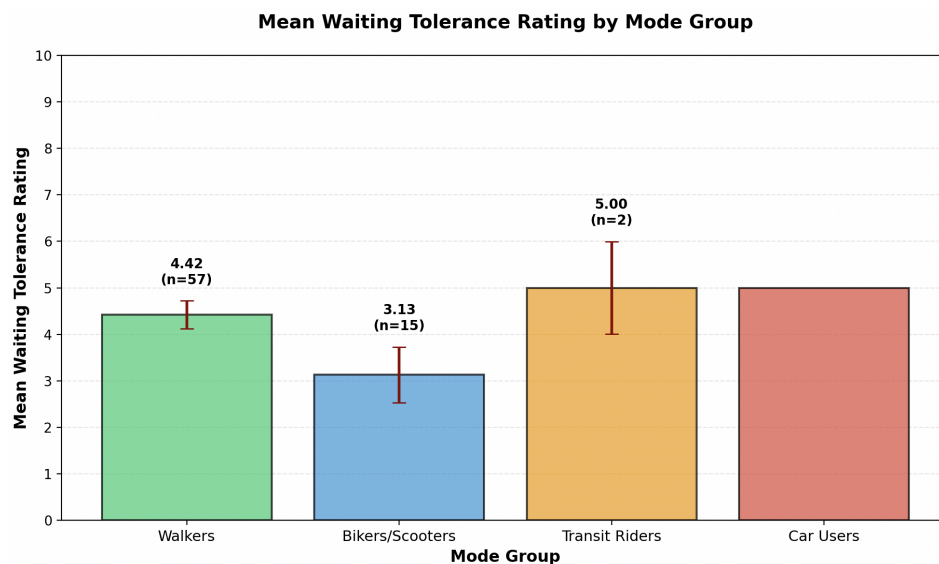


Figure 10. Mean waiting tolerance rating by mode

Again, this aligns with H3, indicating that tolerance emerges from mode rather than from commute distance.



## Consistency Between Numeric and Multiple-Choice Measures

A correlation plot comparing numeric wait tolerance (minutes) with the waiting tolerance rating (0-10) shows a weak but positive relationship.

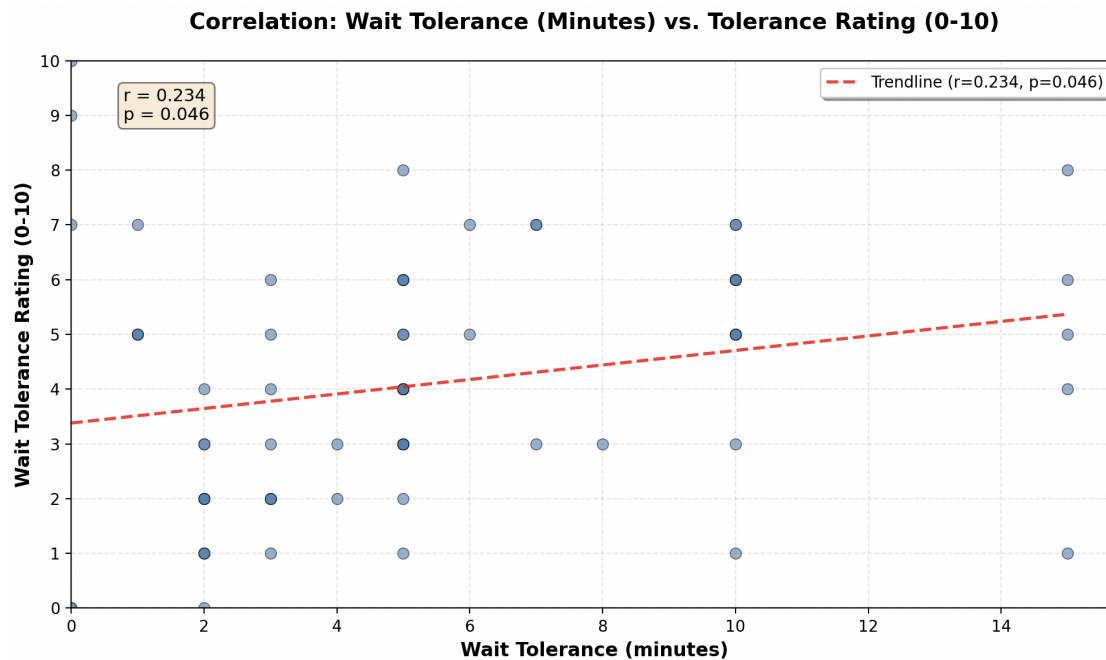


Figure 11. Correlation between wait tolerance (minutes) and wait tolerance rating

This suggests that while the two measures are related, they likely capture different behavioral constructs: one is a behavior-based threshold, the other a general personality or attitude toward waiting.

## Interpretation Summary

- Distance shapes mode choice, not waiting tolerance.
  - Students farther from campus overwhelmingly choose biking or scootering, minimizing exposure to waiting. This supports H2.
- Transit users (not long-distance commuters) have the highest waiting tolerance.
  - This supports H3: waiting tolerance appears to be an adaptive behavior conditioned by transit use rather than a function of commute length.
- Most students avoid waiting whenever possible.
  - Across distance groups, the dominant response to delay was to switch modes immediately, highlighting the strong preference for autonomy and control described in the literature.

## Discussion

### Impact and/or Implications

Although one might expect students living farther from campus to develop greater tolerance for delays given longer overall commute burdens, the results suggest a different behavioral mechanism. Rather than adapting by increasing patience, students with longer commutes appear to adapt by selecting modes that minimize or eliminate exposure to waiting altogether, such as biking or scootering. In contrast, higher waiting tolerance is observed primarily among transit users, consistent with behavioral adaptation to scheduled, headway-based systems rather than differences driven by commute distance itself.

The consistently higher waiting tolerance observed among transit users reflects a different behavioral pattern: adaptation through habituation. Students who rely on MBTA or shuttles may not prefer waiting, but they incorporate it into their routines, plan buffer time, and develop realistic expectations about headways. This insight has practical implications for campus transit service design. For instance:

- Providing real-time information or predictable headways may disproportionately benefit users who already tolerate waiting but need reliability.
- Meanwhile, encouraging mode shifts among short-distance commuters may require addressing the perceived inefficiency of waiting compared to simply walking or biking.

Overall, the findings suggest that campus transportation policy should not assume all commuters value speed or reliability equally. Instead, the value of waiting time differs across mode users, and interventions that reduce uncertainty may matter more than those that reduce waiting duration itself.

## **Limitations**

Several limitations should be considered when interpreting these results. First, the sample was heavily skewed toward on-campus students, limiting our ability to generalize patterns among true long-distance commuters. Only one respondent lived more than three miles away, and thus our highest distance category largely reflects atypical variation. Second, the sample size is small ( $n = 75$ ), reducing statistical power and limiting our ability to detect subtle relationships between variables such as distance and waiting preference. Furthermore, the number of respondents who take the shuttle to Lobby 7 was only 2, which may have heavily skewed results.

Third, the study relies on self-reported wait tolerance rather than observed behavior. Stated preferences often diverge from actual behavior, particularly under time pressure, weather conditions, or competing obligations. Additionally, respondents may have interpreted questions differently: some might have assumed ideal conditions, while others incorporated worst-case delays into their responses.

Finally, mode choice and waiting tolerance are influenced by many contextual factors we did not measure, including bike availability, seasonal weather, prior negative transit experiences, or risk



preferences. Without these controls, it is difficult to isolate pathways linking commute distance, waiting tolerance, and behavioral responses to delays.

## **Extensions and Future Research**

Future research could address these limitations and deepen our understanding of waiting behavior in several ways. One promising direction is to use revealed-preference data, such as real-time observations at campus bus stops or smartphone-based mobility traces, to capture actual switching behavior when delays occur. This would allow researchers to compare stated and revealed wait thresholds and test whether tolerance differs under real-world pressure.

A second avenue involves collecting a more balanced sample of off-campus students, whose commute decisions likely vary more meaningfully in response to distance, seasonal factors, and transit reliability. Stratified sampling across neighborhoods such as Cambridgeport, Somerville, Allston, or Brookline could reveal broader patterns missed in our dataset.

Further research might also examine situational moderators, such as weather, trip purpose, time of day, academic workload, or crowding at stops. These contextual factors likely interact with waiting tolerance in ways that could inform scheduling adjustments or targeted reliability improvements.

Finally, integrating our findings with behavioral interventions such as providing estimated arrival times, encouraging time-use opportunities during waiting, or offering micro-mobility connections could test whether perceived waiting burden can be shifted without altering headways themselves.

## **Conclusion**

This study set out to understand how commute distance and mode choice influence MIT students' waiting tolerance and behavioral responses to delays. Contrary to the initial hypothesis, students living farther from campus do not exhibit greater patience; instead, they avoid waiting by choosing modes that maximize autonomy and reliability. Transit users, however, display distinctly higher waiting tolerance, likely shaped by habitual adaptation to scheduled or headway-based systems.

Across the sample, most students responded to delays not by waiting longer but by switching modes, demonstrating a strong preference for control and predictability. These findings underscore the importance of designing mobility systems that minimize uncertainty rather than simply minimizing wait times.

While constrained by a small and unbalanced sample, this study highlights key behavioral patterns relevant for campus planners and transit operators. Strengthening reliability, offering

flexible alternatives, and acknowledging the diverse tolerance levels of different commuter groups may all help improve the user experience and encourage more sustainable mode choices on campus.

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## Survey Raw Data

### Link to Survey Results:

[https://docs.google.com/spreadsheets/d/1ccvnm7YN-fWi6VE4yqbR\\_q0KPKejL0JsQ3-IUfTmADw/edit?resourcekey=&gid=1207336323#gid=1207336323](https://docs.google.com/spreadsheets/d/1ccvnm7YN-fWi6VE4yqbR_q0KPKejL0JsQ3-IUfTmADw/edit?resourcekey=&gid=1207336323#gid=1207336323)

Link to Survey Responder Form: <https://forms.gle/U3aKPxvkS2HLG6Yp7>

| Timestamp          | Where do you live? | What dorm do you live in? | Approximately how far do you live from campus (in miles)? | What is your primary commute mode?                                                 | What is your typical one-way commute time to Lobby 7 (in minutes) | If you had to wait for a campus shuttle or MBTA bus during your commute, how long (in minutes) would you be willing to wait before switching to another option (e.g., walking, biking, scootering)? | How tolerant are you of waiting for transit? | You're running late and the bus is delayed by 8 minutes. What do you most likely do? |
|--------------------|--------------------|---------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------|
| 12/1/2025 21:58:32 | On-campus dorm     | EC                        |                                                           | Walking                                                                            | 7 min                                                             | 0                                                                                                                                                                                                   | 7                                            | Walk or bike instead (if feasible but takes longer)                                  |
| 12/1/2025 21:58:41 | On-campus dorm     | New House                 |                                                           | Walking                                                                            | 10                                                                | 5                                                                                                                                                                                                   | 2                                            | Walk or bike instead (if feasible but takes longer)                                  |
| 12/1/2025 21:58:51 | On-campus dorm     | New House                 |                                                           | Bike                                                                               | 4                                                                 | 2                                                                                                                                                                                                   | 3                                            | Walk or bike instead (if feasible but takes longer)                                  |
| 12/1/2025 21:59:07 | On-campus dorm     | Maseeh                    |                                                           | Walking                                                                            | 3                                                                 | 0                                                                                                                                                                                                   | 9                                            | Wait for the bus                                                                     |
| 12/1/2025 21:59:27 | On-campus dorm     | Next House                |                                                           | Bike                                                                               | 5                                                                 | 5                                                                                                                                                                                                   | 4                                            | Walk or bike instead (if feasible but takes longer)                                  |
| 12/1/2025 22:00:06 | On-campus dorm     | McCormick                 |                                                           | Walking, I have a non-electric scooter for further (>10 min walk) campus buildings | <5                                                                | 0                                                                                                                                                                                                   | 0                                            | Walk or bike instead (if feasible but takes longer)                                  |

|                       |                                   |               |                                 |         |      |                                                                                                                                                                                                                               |   |                                                              |
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| 12/1/2025<br>22:00:30 | Off-campus<br>within 0-3<br>miles |               | 1.1 from<br>my house<br>to stud | Bike    | 9    | 5                                                                                                                                                                                                                             | 5 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:00:33 | On-campus<br>dorm                 | EC            |                                 | Walking | 8    | it depends on<br>how far i need to<br>go, how urgent<br>my planned<br>arrival time is,<br>and what other<br>transport options<br>are available. but<br>i'd say like 10-20<br>minutes is a<br>decent estimate<br>of most cases | 7 | Wait for the bus                                             |
| 12/1/2025<br>22:00:54 | On-campus<br>dorm                 | Next<br>House |                                 | Walking | 9    | 2                                                                                                                                                                                                                             | 2 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:01:28 | On-campus<br>dorm                 | Simmons       |                                 | Walking | 17   | 7                                                                                                                                                                                                                             | 7 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:03:41 | On-campus<br>dorm                 | Random        |                                 | Walking | 8    | 3                                                                                                                                                                                                                             | 4 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:04:01 | On-campus<br>dorm                 | Next<br>House |                                 | Walking | 8    | 2                                                                                                                                                                                                                             | 2 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:04:05 | On-campus<br>dorm                 | New<br>House  |                                 | Walking | 7ish | 3                                                                                                                                                                                                                             | 2 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:07:25 | On-campus<br>dorm                 | EC            |                                 | Walking | 6    | 7                                                                                                                                                                                                                             | 3 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:11:12 | On-campus<br>dorm                 | Simmons       |                                 | Walking | 10   | 2                                                                                                                                                                                                                             | 2 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:12:10 | On-campus<br>dorm                 | Maseeh        |                                 | Walking | 6-7  | 5                                                                                                                                                                                                                             | 3 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:12:18 | On-campus<br>dorm                 | EC            |                                 | Walking | 5    | 10                                                                                                                                                                                                                            | 1 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:15:38 | On-campus<br>dorm                 | Random        |                                 | Walking | 7    | 1                                                                                                                                                                                                                             | 5 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>22:18:59 | On-campus<br>dorm                 | Maseeh        |                                 | Walking | 4    | 5                                                                                                                                                                                                                             | 4 | Walk or bike<br>instead (if                                  |

|                       |                                   |            |     |         |     |        |    |                                                     |
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|                       |                                   |            |     |         |     |        |    | feasible but takes longer)                          |
| 12/1/2025<br>22:19:06 | On-campus<br>dorm                 | mccormick  |     | Walking | 3   | 5      | 6  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:19:54 | Off-campus<br>within 0-3<br>miles |            | 1   | Bike    | 7   | 0      | 0  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:22:08 | Off-campus<br>within 0-3<br>miles |            | 1.5 | Bike    | 8   | 5      | 6  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:22:23 | Off-campus<br>within 0-3<br>miles |            | 1.1 | Walking | 30  | 5      | 3  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:22:24 | On-campus<br>dorm                 | Next House |     | Bike    | 5   | 1      | 5  | Wait for the bus                                    |
| 12/1/2025<br>22:22:27 | On-campus<br>dorm                 | Simmons    |     | Walking | 10  | 5      | 4  | Wait for the bus                                    |
| 12/1/2025<br>22:25:15 | On-campus<br>dorm                 | Random     |     | Walking | 7   | 15     | 5  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:26:28 | On-campus<br>dorm                 | Maseeh     |     | Walking | 3   | 3      | 2  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:26:55 | On-campus<br>dorm                 | Next House |     | Walking | 10  | 10     | 7  | Wait for the bus                                    |
| 12/1/2025<br>22:27:06 | On-campus<br>dorm                 | Next House |     | Walking | 10  | 3      | 2  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:34:05 | On-campus<br>dorm                 | Random     |     | Walking | 7   | 5      | 4  | Wait for the bus                                    |
| 12/1/2025<br>22:35:33 | On-campus<br>dorm                 | EC         |     | Walking | 10  | 10     | 5  | Wait for the bus                                    |
| 12/1/2025<br>22:35:52 | On-campus<br>dorm                 | Tang       |     | Bike    | 5   | 3      | 3  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:36:25 | On-campus<br>dorm                 | New Vassar |     | Walking | 5   | 5 mins | 4  | Walk or bike instead (if feasible but takes longer) |
| 12/1/2025<br>22:41:29 | Off-campus<br>within 0-3<br>miles |            | 2.7 | MBTA    | 35  | 15     | 6  | Wait for the bus                                    |
| 12/1/2025<br>22:44:09 | Off-campus<br>within 0-3<br>miles |            | 1   | Walking | 11  | 10     | 6  | Wait for the bus                                    |
| 12/1/2025<br>22:44:33 | On-campus<br>dorm                 | New House  |     | Walking | 6.5 | 0      | 10 | Wait for the bus                                    |

|                       |                                         |                       |     |         |           |                                                                  |   |                                                              |
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| 12/1/2025<br>22:53:17 | On-campus<br>dorm                       | BC                    |     | Walking | 7 minutes | 10 minutes                                                       | 6 | Wait for the bus                                             |
| 12/1/2025<br>23:00:58 | On-campus<br>dorm                       | BC                    |     | Walking | 10        | 5                                                                | 3 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>23:02:23 | On-campus<br>dorm                       | BC                    |     | Walking | 8         | 5                                                                | 5 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>23:09:36 | On-campus<br>dorm                       | New<br>House          |     | Scooter | 7         | 2                                                                | 2 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>23:13:27 | Off-campus<br>more than 3<br>miles away |                       | 7   | Car     | 30-35     | 6                                                                | 5 | Call/use<br>rideshare<br>(Uber/Lyft)                         |
| 12/1/2025<br>23:22:39 | On-campus<br>dorm                       | Maseeh                |     | Walking | 6         | 8                                                                | 3 | Wait for the bus                                             |
| 12/1/2025<br>23:27:25 | On-campus<br>dorm                       | Simmons               |     | Walking | 13        | 10                                                               | 5 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>23:49:11 | On-campus<br>dorm                       | EC                    |     | Walking | 10        | < 2min                                                           | 7 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/1/2025<br>23:57:55 | On-campus<br>dorm                       | Random                |     | Walking | 5         | 3                                                                | 5 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>0:11:30  | On-campus<br>dorm                       | New<br>Vassar<br>bruh |     | Walking | 5         | 2                                                                | 1 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>0:24:43  | On-campus<br>dorm                       | Tang Hall             |     | Bike    | 4         | 2                                                                | 1 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>0:43:03  | On-campus<br>dorm                       | Maseeh                |     | Walking | 5         | I cannot bike or<br>scooter so<br>infinite time<br>theoretically | 8 | Wait for the bus                                             |
| 12/2/2025<br>0:47:36  | Off-campus<br>within 0-3<br>miles       |                       | 0.6 | Walking | 20        | 10                                                               | 5 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>0:49:19  | On-campus<br>dorm                       | Maseeh                |     | MBTA    | 3         | 15                                                               | 4 | Wait for the bus                                             |
| 12/2/2025<br>1:23:55  | On-campus<br>dorm                       | Maseeh                |     | Walking | 3         | 10                                                               | 5 | Wait for the bus                                             |
| 12/2/2025<br>2:36:07  | On-campus<br>dorm                       | Pi Beta<br>Phi        |     | Walking | 10        | 6                                                                | 7 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>4:33:49  | On-campus<br>dorm                       | Simmons               |     | Walking | 20        | 10                                                               | 7 | Walk or bike<br>instead (if                                  |

|                    |                             |                        |      |         |         |                                       |   |                                                     |
|--------------------|-----------------------------|------------------------|------|---------|---------|---------------------------------------|---|-----------------------------------------------------|
|                    |                             |                        |      |         |         |                                       |   | feasible but takes longer)                          |
| 12/2/2025 6:43:54  | Off-campus within 0-3 miles |                        | 0.65 | Walking | 15      | 7                                     | 7 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 7:04:55  | On-campus dorm              | New Vassar.....<br>... |      | Walking | 5 mins  | 3 mins                                | 6 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 8:11:08  | On-campus dorm              | McCormick              |      | Walking | 4       | 5                                     | 4 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 8:13:03  | On-campus dorm              | Tang Hall              |      | Walking | 12 mins | 10                                    | 3 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 8:14:12  | On-campus dorm              | Maseeh                 |      | Walking | 6       | 2                                     | 1 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 8:30:43  | On-campus dorm              | Maseeh                 |      | Walking | 2       | 5                                     | 3 | Wait for the bus                                    |
| 12/2/2025 9:03:51  | On-campus dorm              | Maseeh                 |      | Walking | 4       | 2                                     | 3 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 9:31:52  | On-campus dorm              | Tang                   |      | Bike    | 7       | 2                                     | 0 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 9:35:40  | On-campus dorm              | Next House             |      | Walking | 11      | 5                                     | 8 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 9:42:32  | On-campus dorm              | EC                     |      | Walking | 7       | 10                                    | 6 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 9:46:14  | On-campus dorm              | New House              |      | Walking | 7       | 4                                     | 3 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 11:21:01 | On-campus dorm              | BC                     |      | Bike    | 5       | 1                                     | 5 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 11:51:21 | Off-campus within 0-3 miles |                        | 0.5  | Walking | 10      | 2 (if not raining or extreme weather) | 4 | Walk or bike instead (if feasible but takes longer) |
| 12/2/2025 12:09:36 | On-campus dorm              | BC                     |      | Walking | 8       | 4                                     | 2 | Wait for the bus                                    |
| 12/2/2025 12:53:06 | Off-campus within 0-3 miles |                        | 1    | Bike    | 5       | 5                                     | 6 | Walk or bike instead (if feasible but takes longer) |

|                       |                                   |               |     |         |    |    |   |                                                              |
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| 12/2/2025<br>12:56:01 | Off-campus<br>within 0-3<br>miles |               | 0.7 | Walking | 15 | 5  | 1 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>13:20:40 | Off-campus<br>within 0-3<br>miles |               | 1   | Walking | 20 | 5  | 4 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>14:40:22 | On-campus<br>dorm                 | Next<br>House |     | Bike    | 4  | 10 | 6 | Wait for the bus                                             |
| 12/2/2025<br>15:04:47 | On-campus<br>dorm                 | Random        |     | Walking | 5  | 15 | 8 | Call/use<br>rideshare<br>(Uber/Lyft)                         |
| 12/2/2025<br>22:17:30 | On-campus<br>dorm                 | Simmons       |     | Bike    | 5  | 3  | 1 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/2/2025<br>23:56:38 | On-campus<br>dorm                 | MacGreg<br>or |     | Walking | 11 | 15 | 1 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |
| 12/3/2025<br>1:23:50  | On-campus<br>dorm                 | Maseeh        |     | Walking | 3  | 5  | 3 | Walk or bike<br>instead (if<br>feasible but takes<br>longer) |